CHAPTER 23 · THE EXAM ARSENAL

Numbers to Memorise — Master Tables

Every figure, frequency, code and definition the examiner is most likely to demand, pulled together from all 22 chapters into one place. This is your final-week reference: read it, cover a column, and recite. If a number is here, it is worth knowing cold.

CONSOLIDATED FROM MODULES A, B & C

Pair this chapter with the Chapter 24 Question Bank for your final exam preparation.

How to use this chapter

- Each table is a self-test: use the button below to blur the right-hand column, then tap/hover to reveal the value from the cue.
- The "Quick cards" at the end are the reference sheets to photograph and carry.
- Finish with the rapid-fire self-test, then move to the Chapter 24 question bank.

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A · Regulations, bodies & documents

B · Licensing, Annexes & equipment

C · Q-codes, pressure & definitions

D · FIRs, frequencies & AIS

E · Meteorology & RVR

F · Electrical & wave relationships

G · Frequency bands & propagation

H · Modulation & equipment

I · Navigation aids & squawks

J · Phraseology, numbers & procedures

☆ Quick cards

? Rapid-fire self-test

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When in doubt, consult the tables. These are the hard facts and figures you must know for both the written exam and the oral viva.

A Regulations, bodies & documents

Cue	Value
Chicago Convention	Signed 7 Dec 1944 (52 States); in force 4 Apr 1947; created ICAO; 96 Articles, 19 Annexes
Article 1	Complete & exclusive sovereignty over national airspace
Article 38	Notify ICAO of any differences from a Standard
ICAO	UN specialised agency · HQ Montreal · Assembly / Council(36) / ANC / Secretariat
ITU	UN agency · HQ Geneva · founded 1865 (oldest) · ITU-R/T/D · allocates spectrum (Radio Regs)
ITU Regions	Region 1 (Europe/Africa/ME), Region 2 (Americas), Region 3 (Asia-Pacific — India)
IATA	Airline trade association (not a regulator); 3-letter airport / 2-letter airline codes
Standard vs Recommended	Standard = necessary (comply or file difference); Recommended = desirable
India	DGCA (regulator/CARs) · AAI (ANS) · WPC (spectrum/NFAP) · BCAS (security) · AAIB (investigation)

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Cue	Value
Key ICAO docs	Annex 10 (telecom) · Doc 4444 PANS-ATM · Doc 8168 PANS-OPS · Doc 9432 RTF Manual

B Licensing, the 19 Annexes & equipment

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Cue	Value
RTR(A)	Radio Telephone Operator (Restricted), Aeronautical — for the aeronautical mobile service
Two licences	Operator's RTR(A) + aircraft radio station licence (apparatus, via WPC)
Governing law	Bharatiya Vayuyan Adhiniyam 2024 · RTR (Restricted) Rules 2025 · CAR Sec 7, Series G, Part VI
Minimum equipment	ICAO Annex 6 (Operation of Aircraft); scales up for IFR/night
Independence rule	One failure must not disable both comms and navigation
Dispatch u/s	Governed by the MEL (from the MMEL)

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#	Annex	#	Annex
1	Personnel Licensing	**11**	Air Traffic Services
2	Rules of the Air	**12**	Search and Rescue
3	Meteorological Service	**13**	Accident & Incident Investigation
4	Aeronautical Charts	**14**	Aerodromes
5	Units of Measurement	**15**	Aeronautical Information Services
6	Operation of Aircraft	**16**	Environmental Protection
7	Aircraft Nationality & Registration	**17**	Security
8	Airworthiness of Aircraft	**18**	Dangerous Goods
9	Facilitation	**19**	Safety Management
10 ☆	**Aeronautical Telecommunications** — 5 volumes (RTR uses Vol I & II)		

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C Q-codes, pressure & vertical definitions

Cue	Value
QNH	Set → altimeter reads ALTITUDE (AMSL); reads aerodrome elevation on the ground
QFE	Set → reads HEIGHT above aerodrome; zero on the runway
QNE	Standard 1013.2 hPa (29.92 inHg) → FLIGHT LEVEL
QFF	True MSL pressure (meteorological, not flown)
Pressure lapse	≈ 1 hPa per 27 ft near sea level
Transition	Climb: set 1013 at transition ALTITUDE · Descend: set QNH at transition LEVEL
QDM / QDR	Magnetic heading TO / bearing FROM station (QDR = QDM ± 180°)
QUJ / QTE	True bearing TO / FROM the station
QSY / QGH	Change frequency / controlled descent (DF let-down)
Altitude/Height/Elevation/FL	From MSL / from a datum / of a surface point / referenced to 1013.2 hPa

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D FIRs, frequencies & AIS

Cue	Value
Indian FIRs	Delhi, Mumbai, Kolkata, Chennai (+ Chennai Oceanic)
ICAO location codes	VIDP Delhi · VABB Mumbai · VECC Kolkata · VOMM Chennai (India = V, 4 letters)
Airspace classes	A (all controlled) → G (uncontrolled)
VHF ATC band	118.0 – 136.975 MHz (AM, line-of-sight)
Emergency	121.5 MHz (VHF) · 243.0 MHz (UHF mil) · 406 MHz (ELT) · 123.1 (SAR aux)
Maritime distress	2182 kHz (MF)
IAIP	AIP (GEN/ENR/AD) · Amendment · Supplement · AIC · NOTAM
NOTAM types	NOTAMN (new) · NOTAMR (replace) · NOTAMC (cancel); SNOWTAM, ASHTAM
AIRAC	28-day cycle
AFTN priorities	SS Distress · DD Urgency · FF Flight-safety · GG Met · KK Regularity

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E Meteorology & RVR

Cue	Value
METAR / SPECI / TAF	Observed / special observed / forecast
METAR order	Station · Time(Z) · Wind · Vis · Weather · Cloud · Temp/Dew · QNH · Trend
Cloud oktas	FEW 1–2 · SCT 3–4 · BKN 5–7 · OVC 8 (heights in hundreds of ft)
TAF change groups	FM · BECMG · TEMPO · PROB30/40
CAVOK	Vis ≥10 km · no cloud <5000 ft / no CB·TCU · no significant weather
SIGMET / AIRMET	Hazard to all aircraft / to low-level light aircraft
ATIS / VOLMET	One-aerodrome broadcast / en-route multi-aerodrome broadcast
RVR	Runway Visual Range (metres); points: touchdown · midpoint · stop-end
Wind reference	METAR = degrees TRUE; ATC surface wind = degrees MAGNETIC

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F Electrical & wave relationships

Cue	Value
Units	Volt · Ampere · Ohm (Ω) · Watt · Farad · Henry
Ohm's Law	V = I × R
Power	P = V × I = I²R = V²/R
RMS (sine)	RMS = 0.707 × peak · peak = 1.414 × RMS
Aircraft power	28 V DC · 115 V AC 400 Hz
Series / parallel	Series: R adds, same current · Parallel: R reduces, same voltage
Wave relationship	c = f × λ · c = 3 × 10&sup8; m/s · f = 1/T (higher f ⟶ shorter λ)

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G Frequency bands & propagation

Band	Frequency	Propagation / use
LF/MF	30 kHz – 3 MHz	Ground wave — NDB

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Band	Frequency	Propagation / use
HF	3 – 30 MHz	Sky wave — oceanic voice (SSB)
VHF	30 – 300 MHz	Space wave (line-of-sight) — ATC voice, VOR, ILS LOC
UHF	300 MHz – 3 GHz	Line-of-sight — ILS GP, DME, SSR, GPS

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Cue	Value
HF day/night	Higher frequency by DAY, lower by NIGHT (D layer absorbs by day)
MUF / LUF / OWF	Max usable / Lowest usable / Optimum ≈ 85% of MUF
Dead space	Gap between ground-wave range and first sky-wave return
Radio horizon	d (NM) ≈ 1.23 × √h(ft)
Channel spacing	25 kHz and 8.33 kHz
Aviation VHF polarization	Vertical

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Modulation & equipment

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Cue	Value
AM	Amplitude varied; bandwidth = 2 × highest audio; info in sidebands; ~67% power in carrier
FM	Frequency varied; noise-resistant; capture effect (not used for VHF voice)
SSB	One sideband, carrier suppressed; efficient — HF voice
Aviation choice	VHF voice = AM (A3E) · HF voice = SSB (J3E)
Receiver	Superheterodyne (converts to a fixed IF)
Simplex	One frequency, one transmitter at a time (aviation VHF)
Squelch / AVC	Mutes background hiss / holds received audio level steady

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Navigation aids & squawk codes

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Aid	Band	Gives
NDB / ADF	LF/MF 190–1750 kHz	Relative bearing (±5°)

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Aid	Band	Gives
VOR	VHF 108.0–117.975 MHz	Magnetic radial TO/FROM
ILS LOC / GP	VHF 108.10–111.95 / UHF 329–335 MHz	Azimuth / glidepath
Markers	75 MHz	Distance to runway
DME	UHF 962–1213 MHz	Slant range
SSR	1030 ↑ / 1090 ↓ MHz	Identity (A) + altitude (C) + Mode S
GPS	L1 1575.42 MHz	4 satellites for a 3-D fix
Radio altimeter	4200–4400 MHz	True height (AGL)

Squawk	Meaning
7500	Unlawful interference (hijack)
7600	Radio / communications failure
7700	General emergency
7000 / 2000	Conspicuity (VFR) / entering from a non-SSR area

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J Phraseology, numbers & procedures

Cue	Value
Altered numerals	3 = TREE · 5 = FIFE · 9 = NIN-er · decimal = DAY-SEE-MAL
Number rules	Digit-by-digit (call signs, headings, FL, squawk, freq) · hundreds/thousands (altitude, height, cloud, vis)
Time	UTC (Zulu); minutes only, or full hours+minutes if confusion
ROGER / WILCO	Received / will comply (ROGER ≠ yes; AFFIRM/NEGATIVE = yes/no)
Readability	1 unreadable → 5 perfectly readable
Ground suffixes	Delivery · Ground · Tower · Approach · Departure · Director · Control · Information
Always read back	Runway/level/heading/speed clearances, QNH, squawk, frequency changes
Circuit legs	Upwind · crosswind · downwind · base · final
Distress / urgency	MAYDAY ×3 (grave danger) / PAN-PAN ×3 (safety, not immediate)

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Cue	Value
Silence	Impose: SEELONCE MAYDAY · End: DISTRESS TRAFFIC ENDED, SEELONCE FEENEE
Emergency descent	ATC "ATTENTION ALL AIRCRAFT … emergency descent in progress"
Comms failure	Squawk 7600 · continue per last clearance/flight plan · light signals
Wake categories	Super · Heavy (≥136t) · Medium (7–136t) · Light (≤7t)
VDF bearing classes	A ±2° · B ±5° · C ±10° · D worse
TCAS RA	Obey (vertical only), say "TCAS RA" then "clear of conflict"

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Quick cards

Phonetic alphabet

A Alfa	E Echo	I India
B Bravo	F Foxtrot	J Juliett
C Charlie	G Golf	K Kilo
D Delta	H Hotel	L Lima

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M Mike	R Romeo	W Whiskey
N November	S Sierra	X X-ray
O Oscar	T Tango	Y Yankee
P Papa	U Uniform	Z Zulu
Q Quebec	V Victor	

Numerals: 0 ZE-RO · 1 WUN · 2 TOO · 3 TREE · 4 FOW-er · 5 FIFE · 6 SIX · 7 SEV-en · 8 AIT · 9 NIN-er

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Light signals (no radio)

Signal	In flight	On the ground
Steady green	Cleared to land	Cleared for take-off
Steady red	Give way, continue circling	Stop
Green flashes	Return for landing	Cleared to taxi
Red flashes	Aerodrome unsafe — do not land	Taxi clear of landing area
White flashes	Land & proceed to apron	Return to starting point
Red pyrotechnic	Do not land for the time being (notwithstanding previous instructions)	

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BIG MNEMONICS ON ONE LINE

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Bands: "**V**ery **L**ong **M**odems **H**elp **V**arious **U**seful **S**ignals **E**xcel" · Annex 10 = **TEN=TELECOM** · Law = **Act→Rules→CAR** · Squawks **75 alive / 76 no comms / 77 heaven** · AFTN **S**ome **D**ogs **F**ind **G**ood **K**ennels · QDM = **Direction to Mother** · HF **Day high, Night low.**

? Rapid-fire self-test

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Click an option to check — one from each table

1. The Chicago Convention created ICAO and was signed in:

1944

1947

1919

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Answer: 1944.

Signed 7 Dec 1944; in force (ICAO created) 4 Apr 1947.

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2. Annex 10 covers:

Rules of the Air

Aeronautical Telecommunications

Aerodromes

Answer: Aeronautical Telecommunications.

Annex 10 — Aeronautical Telecommunications (TEN = TELECOM).

3. QNE is the standard setting:

1003.2 hPa

1013.2 hPa

760 hPa

Answer: 1013.2 hPa.

1013.2 hPa (29.92 inHg) → flight levels.

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4. The international VHF emergency frequency is:

243.0 MHz

121.5 MHz

406 MHz

Answer: 121.5 MHz.

121.5 MHz (VHF guard); 243.0 UHF military.

5. The AIRAC cycle is:

7 days

14 days

28 days

Answer: 28 days.

Common effective dates every 28 days.

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6. Broken cloud (BKN) is:

1–2 oktas

3–4 oktas

5–7 oktas

Answer: 5–7 oktas.

FEW 1–2, SCT 3–4, BKN 5–7, OVC 8.

7. c = f × λ, where c is:

3 × 10♠ m/s (approx typo for 10^6)

3 × 10&sup8; m/s

300 m/s

Answer: 3 × 10&sup8; m/s.

Speed of radio waves ≈ 3 × 10&sup8; m/s.

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8. HF long-range voice uses:

AM, ground wave

SSB, sky wave

FM, line-of-sight

Answer: SSB, sky wave.

HF = SSB, sky wave; VHF = AM, line-of-sight.

9. An SSR transponder replies on:

1030 MHz

1090 MHz

121.5 MHz

Answer: 1090 MHz.

Interrogate 1030, reply 1090 MHz.

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10. The squawk for radio failure is:

7500

7600

7700

Answer: 7600.

7600 radio failure; 7500 hijack; 7700 emergency.

11. "ROGER" means:

Yes

I have received all your last transmission

I will comply

Answer: I have received all your last transmission.

ROGER = received; WILCO = will comply; AFFIRM = yes.

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12. The distress signal, spoken three times, is:

PAN-PAN

MAYDAY

SECURITE

Answer: MAYDAY.

MAYDAY ×3 = distress; PAN-PAN ×3 = urgency.

⚡ FINAL-WEEK ROUTINE

- Recite each master table by covering the value column — aim for zero misses.
- Carry the phonetic, light-signal and squawk quick cards; glance daily.
- Then hammer the **Chapter 24 question bank** (297 real questions) until 80%+ everywhere.
- Say the phraseology and the oral answers out loud — the practical decides the result.

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